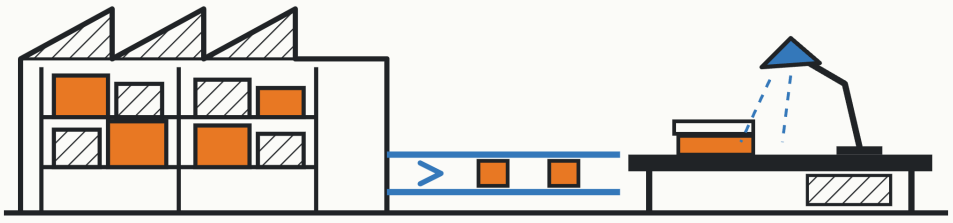
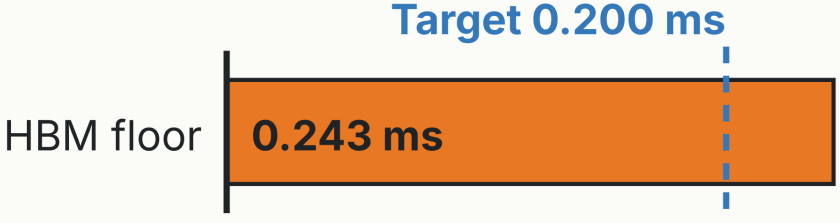
A floor can reject a target

1,856 MiB = 1.946 GB

Assumed 8 TB/s





Reject this dataflow, not every attention design.

Illustrative attention backward

Scratch can become the bottleneck



1,056 MiB unique FP32 partials

128 MiB at most retained between phases

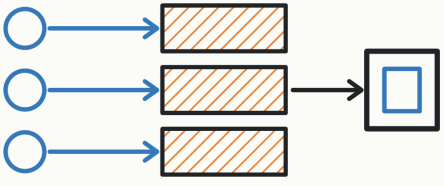
1,856 MiB at least written + reread

One sequence, one layer backward, one device. All production completes before reduction.

Compare the complete path

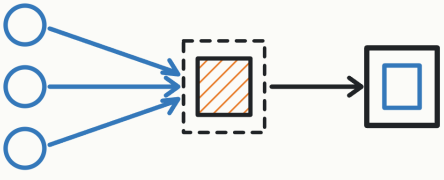
Materialize all

Simple ownership. More scratch and traffic.



Bounded reduction

Less live scratch. More coordination.



Direct accumulation

Less staging. Check synchronization.

